

MP Online AIML Assignment

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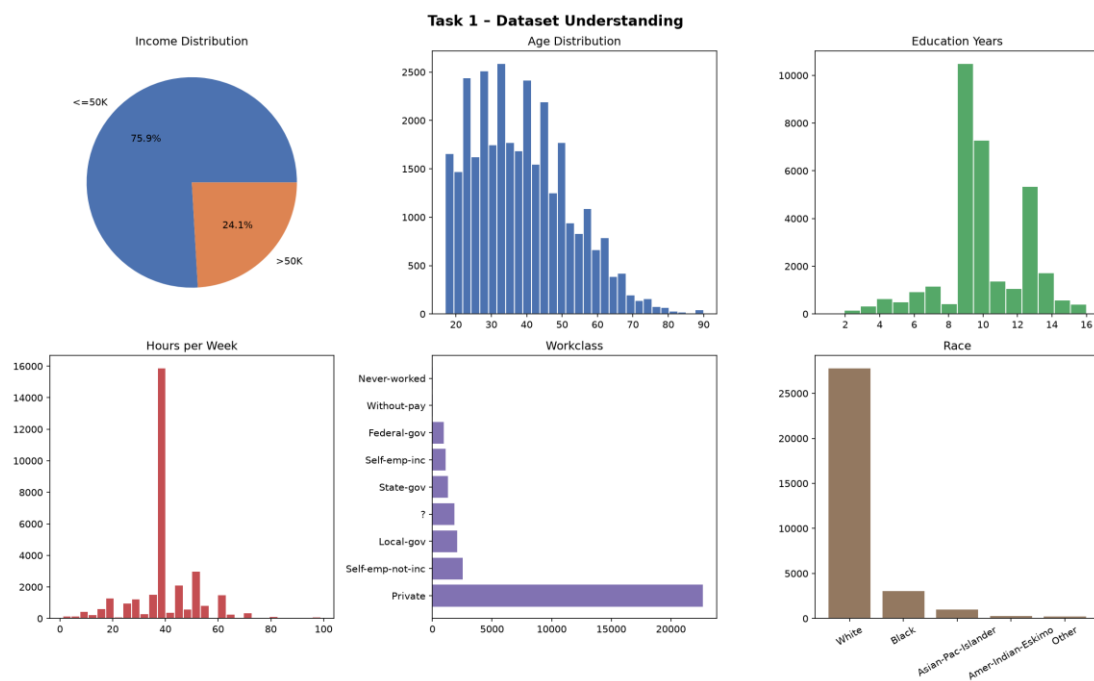
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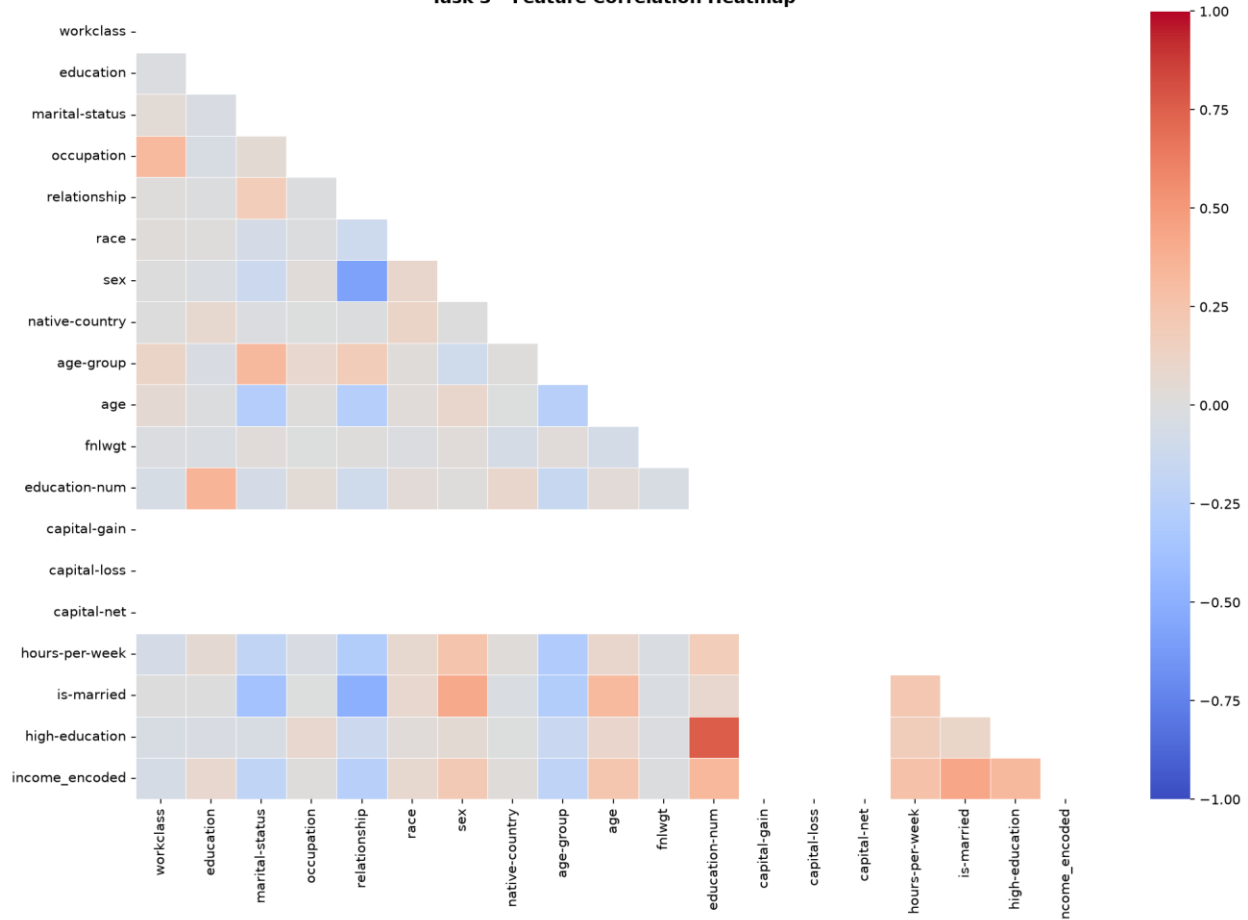
Adult Census Income Dataset

The dataset contains demographic and employment-related information and the target variable is whether an individual's income exceeds \$50,000 per year.

Algorithm	Accuracy	Precision	Recall	F1 Score	ROC - AUC
Logistic Regression	0.8267	0.6800	0.5300	0.5957	0.8737
Decision Tree	0.8254	0.6513	0.5931	0.6208	0.8504
Random Forest	0.8231	0.6512	0.5727	0.6094	0.8645
KNN	0.8161	0.6318	0.5670	0.5976	0.8392
SVM	0.8245	0.7025	0.4713	0.5641	0.8499

Task 1: Dataset Understanding (10 Marks)





Task 4: Model Building (30 Marks) Classification Algorithms

We trained multiple classification models to predict the income brackets:

- **Data Splitting & Scaling:** The dataset was split into an 80% training set and a 20% testing set using stratification to maintain class balance. Numerical features were standardized using StandardScaler for distance-based models.
- **Algorithms Used:** Five models were built and trained:
 - Logistic Regression (baseline linear model)
 - Decision Tree (max depth limited to 10 to prevent overfitting)
 - Random Forest (ensemble of 100 trees for robustness)
 - K-Nearest Neighbors (KNN with k=7)
 - Support Vector Machine (SVM with an RBF kernel for non-linear boundaries)
- **Testing:** Each trained model was used to predict classes and probabilities on the 20% test data, which were then saved for performance evaluation.

Task 5: Performance Evaluation (15 Marks)

